

CS 185/285 Assignment 3 Report

Q-Learning and Actor-Critic Algorithms

2. Deep Q-Learning

2.5 Double Q-Learning

Q. Run double-Q DQN on LunarLander-v2 and plot evaluation return over training steps.

A. Double-Q DQN successfully learned LunarLander-v2 and reached an evaluation return above **200** during training, satisfying the expected performance threshold.

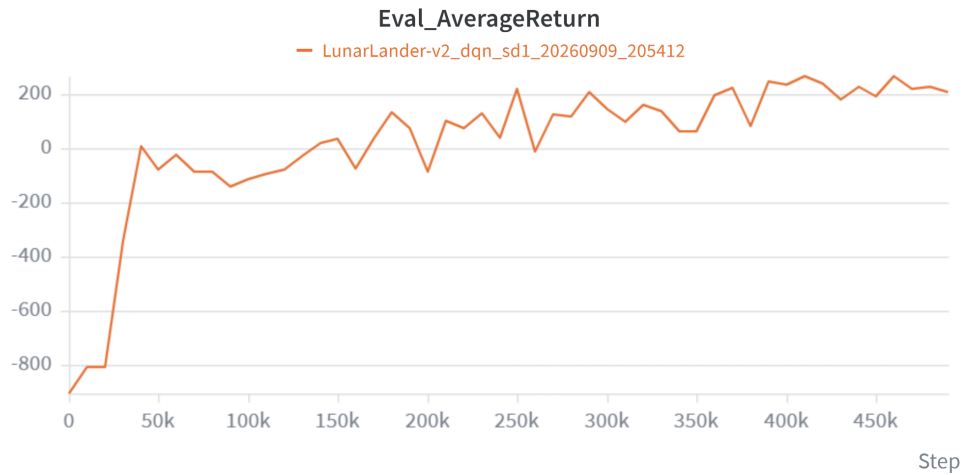


Figure 1. Double-Q DQN evaluation return on LunarLander-v2.

Q. Run DQN on MsPacman-v0. Plot average training return and evaluation return on the same axes. Why do they look different early in training?

A. The MsPacman agent learned steadily and reached roughly **1,500-1,900** evaluation return late in training. Early in training, evaluation return is noisier because training uses an **epsilon-greedy** policy with substantial random exploration, whereas evaluation uses the current Q-network greedily. When the Q-network is still inaccurate, greedy evaluation can repeatedly commit to erroneous action preferences and produce highly variable outcomes. As epsilon anneals, the training policy becomes nearly greedy and the two curves move into a similar range.

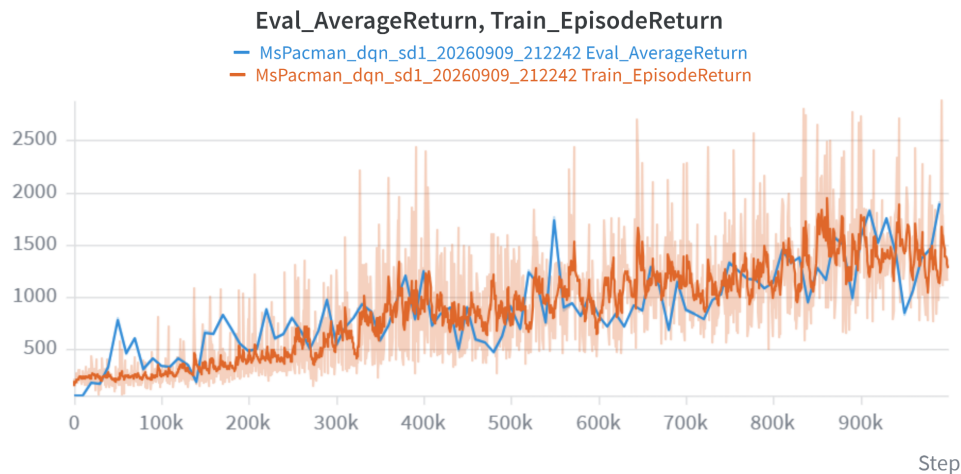


Figure 2. MsPacman training and evaluation returns.

2.6 Experimenting with Hyperparameters

Q. Choose one hyperparameter and run at least three other settings in addition to the previous setting. Plot all four values on the same graph and explain why you chose this hyperparameter.

A. Hyperparameter: target network update period. It was chosen because it directly controls the stability-speed trade-off in Q-learning: very frequent updates can make the bootstrap target move rapidly, while very infrequent updates make learning rely on stale targets. Values tested were **100, 1,000, 5,000, and 20,000** steps, with all other hyperparameters fixed.

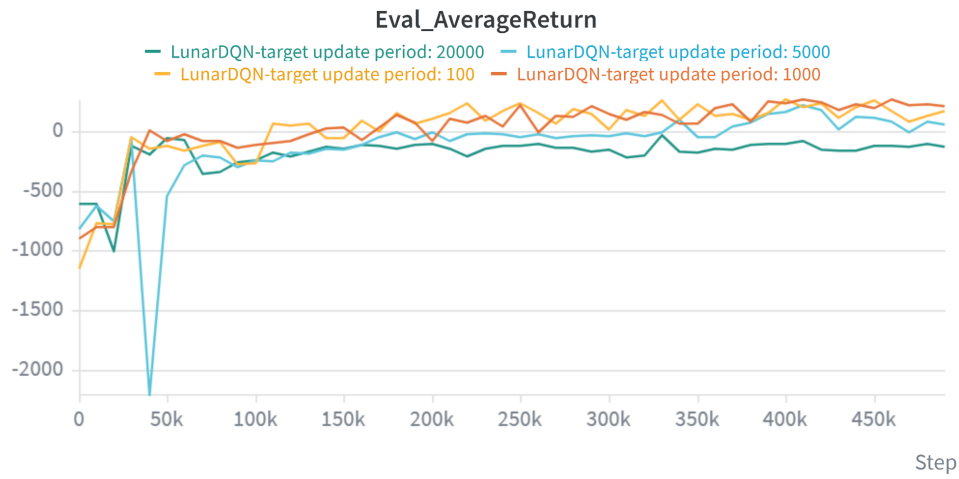


Figure 3. LunarLander-v2 target-network update-period sweep.

A. Periods of **100 and 1,000** performed best and reached positive returns. The **5,000**-step setting showed severe transient instability, dropping below -2,000 around 40k steps before recovering. The **20,000**-step setting performed worst and stayed mostly negative, consistent with targets becoming too stale. Thus, DQN performance is sensitive to the target update period, and shorter periods gave the best trade-off here.

3. Soft Actor-Critic (SAC)

3.4 Actor Update with Reparametrization

Q. Train SAC on HalfCheetah-v4 and plot evaluation return against environment steps. The agent should achieve a return of at least 6000 at least once.

A. The reparameterized SAC actor successfully learned HalfCheetah-v4. Evaluation return increased from below zero to approximately **10,000** by the end of training and exceeded the required return of 6,000.

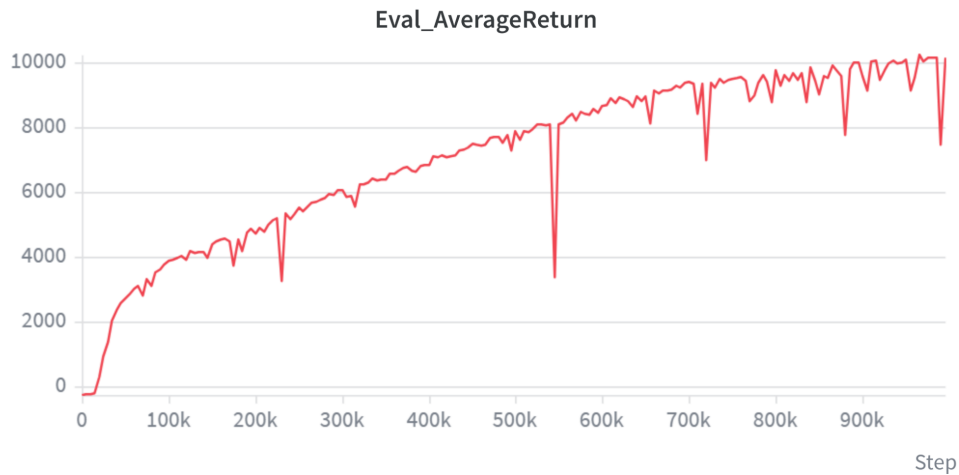


Figure 4. Fixed-temperature SAC evaluation return on HalfCheetah-v4.

3.5 Automatic Temperature Tuning

Q. Create a figure showing evaluation return for fixed and auto-tuned temperatures, and the learned temperature alpha over training.

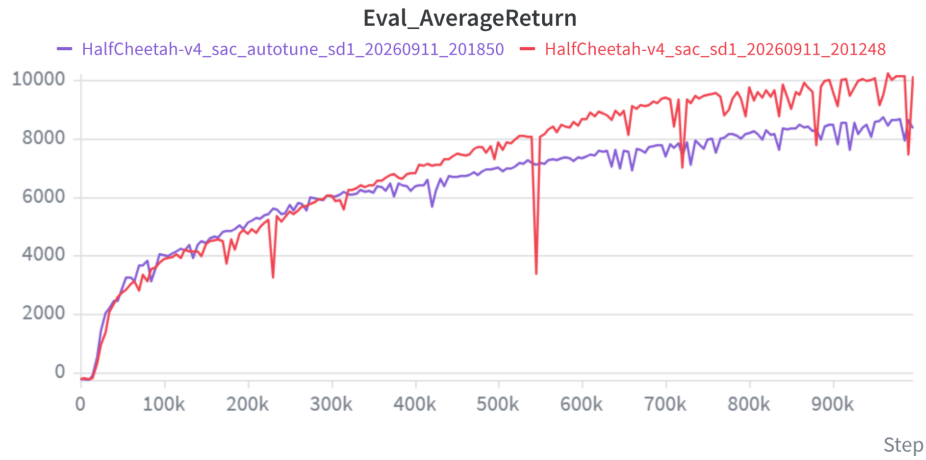


Figure 5. Fixed-temperature vs. auto-tuned SAC on HalfCheetah-v4.

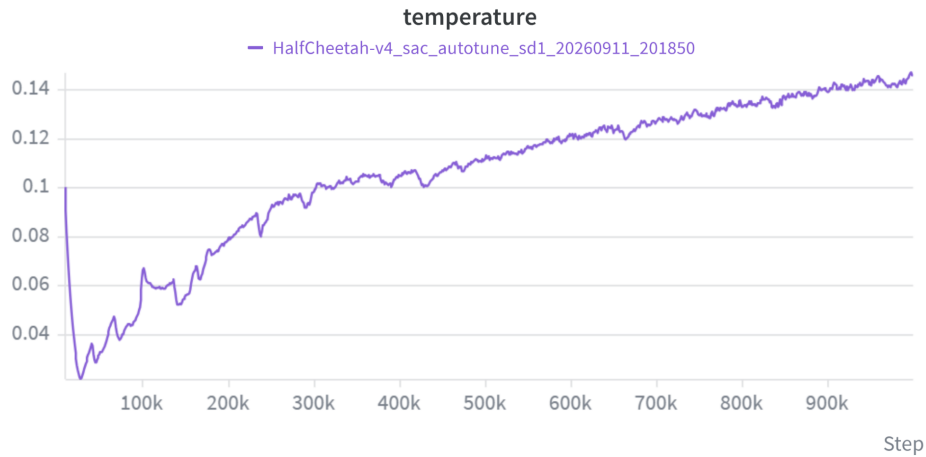


Figure 6. Learned temperature alpha during auto-tuned SAC training.

Q. Does auto-tuning improve or achieve comparable performance to the fixed temperature?

A. Auto-tuning achieves reasonably comparable performance but does not outperform the fixed-temperature run. It reaches approximately **8,500** return, while fixed-temperature SAC reaches approximately **10,000**.

Q. How does the temperature evolve during training? Does it increase, decrease, or stabilize?

A. The temperature initially decreases sharply from **0.1** to about **0.02**, then gradually increases throughout training, reaching approximately **0.145** near the end.

Q. Why might the temperature change in this particular way for HalfCheetah?

A. HalfCheetah has a 6-dimensional action space, so the target entropy is set to **-6**. Alpha is adjusted to keep policy entropy near this target. It initially decreases when less entropy regularization is needed, then increases as the policy becomes more deterministic, strengthening the entropy bonus to encourage exploration and maintain the target entropy.

3.6 Stabilizing Target Values

Q. Run single-Q and clipped double-Q on Hopper-v4. Which works best? Plot evaluation return and Q-values, and discuss how the results relate to overestimation bias.

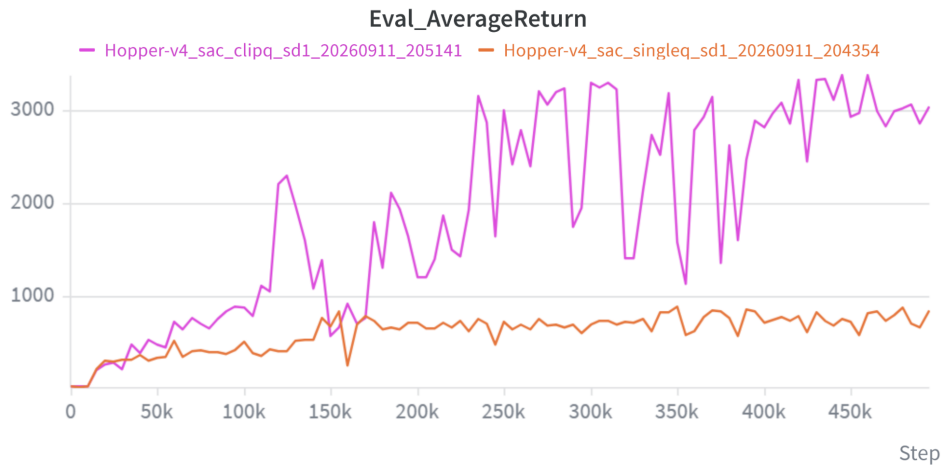


Figure 7. Hopper-v4 evaluation return: single-Q vs. clipped double-Q.

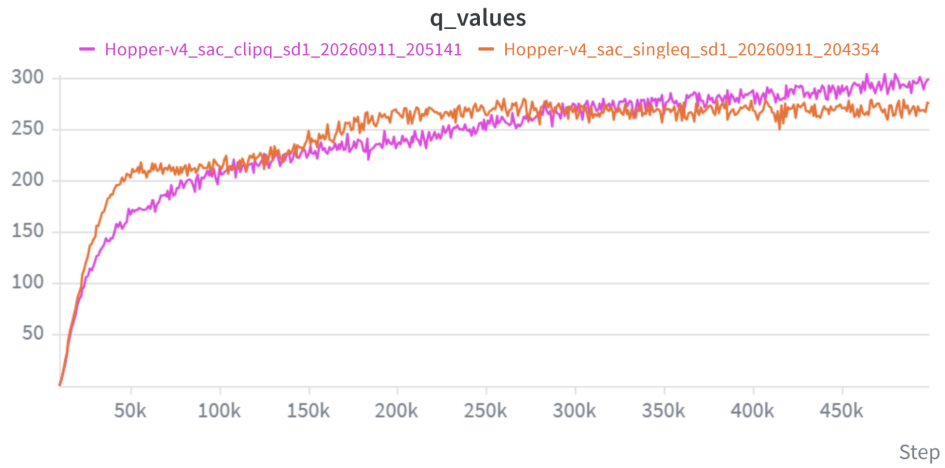


Figure 8. Logged Q-values: single-Q vs. clipped double-Q.

A. The **clipped double-Q** method works much better, reaching returns above **3,000**, while single-Q plateaus around **700-800**. Single-Q often estimates Q-values as high as or higher than clipped double-Q despite its much lower actual return, indicating **overestimation bias**. Clipped double-Q uses $\min(Q1, Q2)$ for the backup, reducing positive estimation errors and producing more reliable targets, which leads to substantially better policy performance.